CSE 5104 Project 1
Concrete Compressive Strength Analysis Part 2: Client Report

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Executive Summary

A ready-mix producer needs to balance material use, production schedules, and dependable concrete strength. We analyzed concrete mixtures to identify useful patterns and build a model that can help compare proposed batches. Our main finding is that strength depends on the combination of ingredients and curing age, so changing one ingredient without considering the rest can be misleading. We recommend using the model to shortlist trial mixtures and plan testing, with measured strength remaining the basis for accepting a batch.

The best-performing model in this comparison used logarithmic transformations of the ingredient amounts and age. It explained about 79.8% of the strength variation in the training data and 75.7% in the separate testing data. The testing result means the model captured about three quarters of the observed differences in strength; it does not mean that 75.7% of its individual predictions were correct. Its root mean squared prediction error was about 7.37 MPa, which is large enough to matter when a mixture is close to a required strength level.

The main recommendations are:

- Control water additions and assess workability together with strength.
- Compare cement and supplementary material choices through small, controlled trials.
- Account for curing age when scheduling tests and interpreting results.
- Track prediction errors on local batches before relying on the model routinely.

Problem Context & Data Overview

For this report, the intended client is a ready-mix producer seeking fewer unsuccessful trial batches and better information for purchasing and scheduling. The dataset contains 1,030 observations with eight inputs and measured compressive strength. Following the project instructions, we used rows 501–630 as the 130-sample testing set and the remaining 900 observations for training. The observations were kept in their original order, and the testing samples were not used to estimate model coefficients or scaling values. The dataset is provided by the UCI Machine Learning Repository [1].

Cement and water form the paste that binds the mixture, while coarse and fine aggregates provide its main solid framework [2]. Blast furnace slag and fly ash are supplementary cementitious materials that can replace part of the cementitious content and influence strength development. Superplasticizer helps concrete remain workable with less water, making it relevant when a producer wants to avoid adding water simply to improve placement [3]. Age records how many days the concrete has cured before strength is measured. These ingredients and age should therefore be considered together when comparing recipes.

Table 1 summarizes the training observations only. Ingredient quantities are in kilograms per cubic meter, age is in days, and strength is in megapascals. The wide ranges show that this dataset includes substantially different mixtures and testing ages. A mean mixture should not be interpreted as an approved recipe, because averages across different designs may not describe a practical batch.

Table 1: Descriptive statistics for the 900 training samples. All ingredients are in kg/m^3; age is in days and strength is in MPa.

Variable	Mean	SD	Min	Max
Cement	277.87	106.00	102.00	540.00
Slag	76.11	84.08	0.00	305.30
Fly ash	57.38	64.26	0.00	200.10
Water	180.72	22.28	121.80	247.00
Superplasticizer	6.86	5.98	0.00	32.20
Coarse aggregate	973.76	77.57	801.00	1145.00
Fine aggregate	772.33	81.91	594.00	992.60
Age	46.04	61.32	1.00	365.00
Strength	36.93	16.65	2.33	82.60

Figure 1 shows the training relationships between cement, age, and strength. Higher cement quantities generally accompany higher strength, but the spread at similar cement levels is substantial. The age plot also shows why age alone is insufficient: mixtures tested at the same age can have very different strengths. These patterns support comparing complete mixtures rather than ranking batches using just one ingredient.

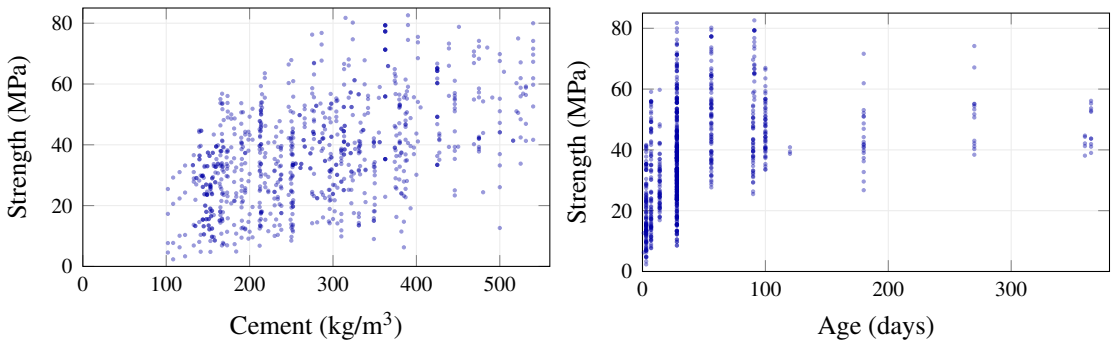


Figure 1: Cement and curing age versus measured strength in the 900 training observations. Points may overlap where measurements share similar values.

Model Reliability & Analytical Findings

Table 2 compares the three fitted models. Rescaling the original inputs changes how the fitting calculation behaves but leaves the final predictions effectively unchanged. Using logarithms changes the relationships the model can represent and produces considerably better results here. The original model explains 62.4% of training variation but only 36.9% of testing variation, whereas the logarithmic model achieves 79.8% and 75.7%, respectively. Its smaller performance drop is encouraging, although one fixed testing sample cannot establish reliability for every plant or supplier.

Table 2: Performance on training and testing samples. Explained variation is expressed as a percentage; RMSE is in MPa.

Model	Train R^2 (%)	Test R^2 (%)	Test RMSE
Raw	62.4	36.9	11.88
Standardized	62.4	36.9	11.88
Log	79.8	75.7	7.37

The original model provides a simple illustration of ingredient effects. Holding the other recorded inputs constant, an additional 10 kilograms of cement per cubic meter is associated with approximately 1.32 MPa higher predicted strength. An additional 10 kilograms of water is associated with approximately 1.34 MPa lower predicted strength. These are comparisons within the fitted data, not guaranteed gains from physically modifying a mixture. A practical redesign must also satisfy volume, workability, and material compatibility constraints.

For the preferred logarithmic model, the estimated change depends on the starting quantity. Increasing cement from 300 to 330 kilograms per cubic meter, with other inputs unchanged, corresponds to about 2.24 MPa higher predicted strength. Increasing water from 180 to 190 corresponds to about 2.06 MPa lower predicted strength. Moving from 7 to 28 days corresponds to about 11.64 MPa higher predicted strength, while moving from 28 to 56 days adds about 6.11 MPa. This pattern allows the benefit of extra curing time to become smaller at later ages.

Water and superplasticizer are closely connected in the training mixtures: batches with more superplasticizer tend to contain less water. This helps explain why a simple comparison of high- and low-superplasticizer batches may also reflect water differences. The fitted models include the ingredients separately; they do not directly estimate a water-to-cement ratio effect or an interaction between water and admixture dosage. We therefore view the results as support for joint water and workability trials, rather than evidence for one universally best ratio.

Strategic Recommendations & Implementation Plan

We propose four operating rules for a limited pilot. First, record all planned and actual water additions, including adjustments during batching, and compare them with the approved recipe. If workability is inadequate, have the quality team assess a controlled superplasticizer adjustment instead of automatically increasing water. The model's negative water association makes uncontrolled additions a reasonable pri-

ority for investigation, but allowable dosages and acceptance decisions must follow the client's established procedures.

Second, compare feasible cement and supplementary material options at the same target age. For example, the quality team can screen a proposed cement change from 300 to 330 kilograms per cubic meter against a lower-cement alternative, then test both under comparable conditions. The purchasing team should add actual supplier prices before deciding whether the estimated strength benefit justifies the expense. The supplied dataset contains no prices, so this analysis cannot identify the cheapest mixture or quantify cost savings.

Third, use a consistent testing schedule when comparing designs. For mixtures intended for 28-day assessment, record earlier measurements separately and avoid treating a 7-day observation as directly comparable with a 28-day result. The age findings suggest that extra curing time can help, but they do not establish when a particular structure can be loaded or released. Those decisions require the appropriate measured results and project requirements.

Fourth, run the model alongside the existing quality process before using it to guide routine planning. For every pilot batch, retain the ingredient amounts, supplier identifiers, curing conditions, test age, predicted strength, and measured strength. Review the differences weekly, checking both the typical error and whether predictions consistently overestimate strength. Review results separately by age group and mixture family so that acceptable overall performance does not hide a problem in a particular product.

The main limitations are missing environmental information, possible differences in local aggregates, and the model's simplified description of ingredient relationships. We would suspend model-based screening for a new supplier or an unfamiliar mixture until representative local tests are available. After enough new batches have accumulated, the quality team should compare performance with the initial 7.37 MPa testing error and investigate any sustained deterioration. Any revised model should be evaluated on newly held-out batches before adoption, with a record of the version used for each prediction.

References

[1] I.-C. Yeh. *Concrete Compressive Strength*. UCI Machine Learning Repository, 1998. https://doi.org/10.24432/C5PK67.

[2] American Concrete Institute. *Cementitious Material in Concrete*. https://www.concrete.org/topicsinconcrete/topicdetail.aspx?search=Cementitious+Material+in+Concrete. Accessed October 14, 2026.

[3] American Cement Association. *Applications of Cement*. https://www.cement.org/cement-concrete/applications-of-cement/. Accessed October 14, 2026.

AI Disclosure

We used ChatGPT (OpenAI) to help us understand coding approaches and troubleshoot errors. We also used it to clarify statistical concepts.Besides, ChatGPT was used to assist with language editing ,and improving the organization and clarity of the report. The analysis and conclusions were reviewed by the project team.